Lydia Plante

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github.com/lydiamplante

Education

CLEMSON UNIVERSITY — Clemson, SC
***Bachelor of Science* in Mechanical Engineering** — May 2029
Minor: Music — 3.44/4.00
First Semester Junior

Undergraduate Projects

NASA Micro-g NExT Creative Inquiry — Clemson, SC
Design and Manufacturing Team — August 2026 - Present

- Objective: Design, build, and test a tool or device that addresses an authentic, current space exploration challenge outlined by NASA
- Research, design, and develop a prototype to put forth a proposal

Aquarium Controller Project — Charleston, SC
Personal Project — June 2026 - Present

- Objective: Design and build an Arduino-based aquarium controller to monitor and automate key tank parameters in a marine tank environment
- Developing foundational skills in Arduino programming, Python, HTML, CSS, and JavaScript while planning the project's hardware and control systems

Leadership & Involvement

Clemson University Symphony Orchestra — Clemson, SC
Concertmaster — August 2024 - Present

- Collaborate with director on orchestral programming to challenge students and engage audiences
- Communicate decisions with violin section and between other section leaders
- Lead execution across the orchestra of 80 musicians

Clemson University Rocket Engineering Club — Clemson, SC
Aerostructures, Simulations & Recovery Team Member — August 2025 - Present

- Created designs for airframe force experiments, fin and nose cone geometries, manufacturing of carbon-fiber and fiberglass airframes, and recovery bay organization under size and electronics organization constraints
- Produced manufacturing ready engineering drawings with tolerances and third-angle projection for Structures and Recovery sub teams
- Attended the ESRA International Rocket Engineering Competition with 12 other selected team members

Work Experience

Cook Mechanical Engineering Student Shop — Clemson, SC
Supervisor — January 2026 - Present

- Instructed and certified 40+ students in the safe and effective operation of mechanical engineering lab equipment, serving 100+ student users per semester across lathes, mills, band saws, MIG welders, and other machining tools
- Planned, proposed, and executed shop improvement projects

Technical Skills & Languages

Engineering Tools: SolidWorks (Proficient), Onshape (Intermediate), Microsoft Office (Proficient), Abaqus (Basic)
Programming: MATLAB (Intermediate), Python (Basic), HTML/CSS/JavaScript (Basic), RStudio (Basic)
Manufacturing: Machine Shop (Proficient), Welding (Proficient), Composite Manufacturing (Intermediate)
Languages: Spanish (B2+ reading, conversational speaking)